BLOOD BANK MANAGEMENT SYSTEM

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

Use Case Report

submitted by

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Kanuru, Vijayawada-520 007

2026-27

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CERTIFICATE

This is to certify that the Use Case report entitled **"Blood Bank Management System"** that is being submitted by **Borra Hema Priya(23501A0518)**, as part of Assignment-2 for the **Blockchain Technology(23CS4701B)** course in **4-1** during the academic year **2026-27**.

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1. INTRODUCTION

1.1. Overview of Blockchain Technology

Blockchain technology is a decentralized and distributed ledger system that records transactions across multiple nodes, ensuring transparency, security, and immutability. Unlike traditional databases, blockchain reduces dependence on a central authority by maintaining a shared record of transactions. Each block contains transaction data and a cryptographic hash linking it to the previous block, making unauthorized modifications detectable. Smart contracts automatically execute predefined rules when specified conditions are met. Initially introduced through cryptocurrency, blockchain technology is now used in finance, healthcare, supply chain management, and digital identity systems. In healthcare, it supports reliable recordkeeping, traceability, and secure information sharing among authorized participants.

1.2. Relevance of Blockchain in Blood Bank Management Systems

Traditional blood bank management systems rely on paper-based records or centralized databases to manage donor details, blood collection, screening results, inventory, and hospital requests. These systems may face problems such as inaccurate records, data loss, unauthorized modifications, and delays in information sharing. During emergencies, identifying available blood supplies and coordinating distribution between blood banks and hospitals can be challenging. Therefore, maintaining accurate and traceable records is essential for efficient blood management.

A blockchain-based blood bank management system provides a secure and verifiable way to record blood-related transactions, from donor registration and collection to laboratory screening, storage, hospital requests, transportation, and receipt. Smart contracts can enforce rules for screening verification, inventory management, request approval, and blood distribution. This reduces manual errors, prevents duplicate allocation of blood units, and improves coordination between blood banks and hospitals. Cryptographic hashing can also help verify the integrity of off-chain records and detect unauthorized changes.

Furthermore, blockchain enables authorized participants to trace blood unit movements and verify transaction histories. Role-based access control restricts operations according to user permissions, while sensitive donor information can be stored securely off-chain. The proposed system aims to improve inventory monitoring, emergency blood requests, inter-hospital transfers, and transaction auditing. Although blockchain cannot independently guarantee blood quality or medical compatibility, integrating it with proper laboratory screening and secure database management can improve the transparency, reliability, and efficiency of blood bank operations.

2. BACKGROUND

Traditional blood bank management systems face several challenges that can affect the safety, availability, and timely distribution of blood. These challenges arise due to the use of paper-based records, centralized databases, and manual processes for managing donors, blood collection, laboratory screening, inventory, and hospital requests. The following are some key issues with existing blood bank management systems:

2.1. Inaccurate or Lost Blood Bank Records

Traditional systems may rely on manual registers and centralized databases to maintain donor details, blood group information, screening results, and inventory records. Human errors, accidental data loss, and outdated information can affect the accuracy of records and create difficulties in managing blood supplies.

2.2. Lack of Blood Unit Traceability

Tracking a blood unit from collection to storage, transportation, and final receipt can be challenging when records are maintained separately by different organizations. Poor traceability makes it difficult to identify the current location and status of a blood unit and verify its movement through the supply chain.

2.3. Delays in Blood Request Processing

Hospitals may need blood urgently during accidents, surgeries, and other medical emergencies. Manual request verification and communication between hospitals and blood banks can cause delays in identifying available units and arranging their distribution.

2.4. Lack of Transparency and Information Sharing

Blood banks and hospitals may maintain separate records that are not updated or shared in real time. This can lead to inconsistent inventory information, confusion regarding request status, and difficulties in coordinating blood supply between healthcare organizations.

2.5. Duplicate Allocation of Blood Units

When inventory records are not updated promptly, the same blood unit may be mistakenly allocated to multiple hospital requests. Such errors can create confusion, waste valuable time, and affect the efficient distribution of available blood supplies.

2.6. Security Risks and Unauthorized Data Modification

Centralized databases and poorly protected records may be vulnerable to unauthorized access or modification. Changes to donor information, screening results, inventory details, or transaction records can reduce trust in the system. Therefore, secure record management and verification mechanisms are essential.

2.7. Difficulties in Monitoring Blood Inventory

Blood banks must continuously monitor available blood groups, unit quantities, screening status, storage conditions, and expiry dates. Inadequate inventory management can result in shortages of required blood groups, unnecessary wastage of expired units, and difficulties in maintaining an appropriate blood supply.

2.8. Challenges in Blood Transportation and Receipt

Coordinating the transportation of blood units from blood banks to hospitals requires accurate information about dispatch, destination, and receipt. Incomplete or delayed updates can make it difficult to confirm whether a unit has reached the intended hospital and whether the transaction has been completed.

2.9. Limited Accountability and Transaction Auditing

In conventional systems, identifying who approved a blood request, modified an inventory record, or dispatched a blood unit may require checking multiple records. Maintaining a reliable history of activities is important for auditing, resolving disputes, and improving accountability among authorized participants.

2.10. Challenges in Emergency Blood Supply Coordination

During emergencies, hospitals need quick access to reliable information about available blood units and their status. Delayed communication and fragmented records can make it difficult to identify suitable supplies and coordinate their timely distribution. Efficient information sharing is therefore important for improving emergency response.

Due to these challenges, a secure and transparent blood bank management system is needed. Blockchain maintains verifiable transaction records, while smart contracts automate blood request approval, unit reservation, and distribution. The proposed Blockchain-Based Blood Bank Management System aims to improve traceability, security, and coordination between blood banks and hospitals. Proper laboratory screening is still essential to ensure blood safety.

3. BLOCKCHAIN BASICS

Blockchain is a distributed ledger technology that records transactions securely and transparently. It uses cryptographic techniques to protect records and allows authorized participants to verify transaction history. In a blood bank management system, blockchain can improve the traceability and reliability of blood-related records. The following are the key concepts of blockchain technology.

3.1. Decentralization

Decentralization allows transaction records to be maintained and verified by multiple nodes in a blockchain network rather than depending entirely on a single central authority. This can reduce dependence on a single point of control and improve the reliability of transaction records. In the proposed blood bank system, blockchain supports the tracking of blood units across different stages of the supply chain. However, the actual level of decentralization depends on the blockchain network on which the application operates.

3.2. Immutability

Immutability refers to the difficulty of changing previously recorded blockchain transactions without detection. Each block is linked to earlier blocks through cryptographic hashes, helping protect the integrity of the transaction history. In this project, blockchain records can help preserve the history of blood unit registration, screening updates, requests, reservations, dispatch, and receipt. If an unauthorized change is attempted, verification mechanisms can help identify inconsistencies. This improves the reliability and auditability of blood bank records.

3.3. Transparency

Transparency allows authorized participants to verify recorded transactions and follow the progress of blood units. Instead of depending entirely on separate manual records, blood banks and hospitals can use the system to check relevant transaction details and request statuses. This can improve communication, reduce discrepancies in inventory information, and support timely coordination during emergencies. Sensitive donor information should remain protected, with access limited according to the application's permissions.

3.4. Smart Contracts

Smart contracts are self-executing programs stored on a blockchain that automatically perform predefined actions when specified conditions are met. They reduce manual work, improve efficiency, and ensure that rules are applied consistently. In blood bank management, smart contracts can automate blood request approval, blood unit reservation, inventory updates, and distribution tracking. They can also help prevent duplicate allocation and maintain a reliable record of transactions. However, medical screening, blood quality, and compatibility must still be verified through proper laboratory procedures and professional medical judgment.

3.5. Consensus Mechanisms

Consensus mechanisms allow nodes in a blockchain network to agree on the validity and ordering of transactions. Different networks use different mechanisms, such as Proof of Work or Proof of Stake, to achieve agreement. The consensus mechanism used by this project depends on the blockchain network selected for deployment; it is not independently implemented by the Solidity smart contract. Consensus helps maintain a consistent transaction history across participating nodes and supports the reliability of blockchain records.

3.6. Cryptographic Security

Cryptographic techniques help protect transaction integrity and verify information recorded by the system. Hash functions convert data into fixed-length values that can be used to detect changes in the original data. The project describes SHA-256 hashes for checking the integrity of off-chain records against their recorded references. If a database record changes unexpectedly, hash verification can help identify the discrepancy. The system also uses role-based access controls to restrict operations according to user permissions. Sensitive donor information is maintained off-chain to reduce unnecessary exposure of personal data.

By combining these blockchain concepts with smart contracts, the proposed Blockchain-Based Blood Bank Management System aims to improve blood unit traceability, record integrity, accountability, and coordination between blood banks and hospitals. It supports more reliable management of requests, reservations, and distribution activities. However, blockchain records alone cannot guarantee blood quality, proper storage temperatures, or medical compatibility. These aspects must be ensured through laboratory screening, suitable storage procedures, and professional medical verification.

4. USE CASE OVERVIEW

The use case for a Blockchain-Based Blood Bank Management System aims to improve traditional blood bank operations by leveraging blockchain technology. The system supports secure, transparent, and traceable management of blood units, from collection and screening to inventory, hospital requests, transportation, and receipt. Smart contracts help automate essential operations and maintain reliable transaction records.

4.1. Objectives

The primary objective of the proposed system is to maintain accurate and reliable records of blood units and their movement through the supply chain. Traditional blood bank systems may rely on manual records or centralized databases, which can lead to errors, outdated information, and difficulties in tracking blood availability.

Another objective is to improve blood unit traceability. Blockchain maintains a verifiable history of blood-related transactions, helping authorized participants track the movement of blood units between blood banks and hospitals. This improves accountability and helps reduce discrepancies in records.

The system also aims to streamline hospital blood requests. Smart contracts can manage request approval, reserve available blood units, and record dispatch and receipt activities according to predefined rules. This helps reduce manual effort and the risk of allocating the same blood unit to multiple requests.

Security and access control are also important objectives. Role-based permissions restrict operations according to the user's responsibilities, while cryptographic techniques help verify record integrity. Sensitive donor information should be protected through appropriate access controls and secure storage.

Furthermore, the system aims to improve inventory monitoring and coordination between blood banks and hospitals. Reliable information about blood unit availability and request status can support better planning and faster responses during emergencies. However, blood quality and compatibility must still be verified through proper laboratory screening and medical procedures.

Additionally, the system aims to maintain a clear transaction history for auditing and verification. It can help identify delays in blood distribution and improve accountability among participating organizations. By reducing paperwork and improving information sharing, the system supports more efficient blood bank operations. These improvements can contribute to better resource utilization and timely availability of blood during emergencies.

4.2. Scope of the System

The Blockchain-Based Blood Bank Management System focuses on recording, tracking, and managing blood supply activities through blockchain and a web-based application. The system involves the following participants and components:

1. **Donors:** Provide the information required for donor registration and blood donation, subject to applicable eligibility and screening procedures.
2. **Blood Banks:** Manage blood unit records, collection details, screening status, inventory, and requests received from hospitals.
3. **Hospitals:** Submit blood requests, check request status, and receive blood units allocated to them.
4. **Administrators:** Manage authorized users and oversee system operations according to assigned permissions.
5. **Blockchain Network:** Maintains verifiable transaction records and supports smart contract execution.
6. **Off-Chain Database:** Stores sensitive or detailed information that should not be placed directly on the blockchain, with integrity checks where supported.

4.3. System Architecture

The system architecture consists of three major layers: the User Layer, Web Application Layer, and Blockchain Layer. These layers work together to support blood bank operations and maintain reliable transaction records.

4.3.1. User Layer

This layer consists of the participants who interact with the system.

- **Donors**:
  - Provide the information required for registration and donation.
  - Support the maintenance of accurate donor records.
- **Blood Banks**:
  - Register and manage blood unit details.
  - Update screening status and inventory information.
  - Process hospital requests and record dispatch activities.
- **Hospitals**:
  - Submit requests for required blood units.
  - Track request progress and allocated units.
  - Confirm receipt of dispatched blood units.
- **Administrators**:
  - Manage user access and permissions.
  - Monitor system activities and support accountability.

4.3.2. Web Application Layer

This layer provides the interface through which authorized users interact with the system.

- **User Interface**:
  - Provides access to registration, inventory, and request-management functions.
  - Allows users to view relevant blood unit details and transaction statuses.
- **Authentication and Access Control**:
  - Verifies user identity.
  - Restricts operations according to assigned roles and permissions.
- **Application Logic**:
  - Processes user requests and coordinates communication between the web application, database, and blockchain.
  - Displays relevant inventory and transaction information.

4.3.3. Blockchain Layer

This layer provides security, transparency, and traceability for blood bank transactions. It maintains verifiable records of blood unit activities and supports the execution of smart contracts.

- **Smart Contract Layer**:
  - **Blood Supply Management Contract**:
    - Manages blood unit details, screening status, inventory, and hospital requests.
    - Supports blood unit reservation and helps prevent duplicate allocation.
    - Controls dispatch and receipt operations according to predefined rules.
  - **Smart Contract Execution**:
    - When a blood unit is registered, its relevant details are recorded according to the contract's rules.
    - When a hospital request is approved, an available unit can be reserved for that request.
    - When a unit is dispatched and received, the corresponding transaction status is updated.
- **Blockchain Network**:
  - **Distributed Ledger**: Maintains a verifiable history of blood supply transactions, helping authorized participants track blood unit movements.
  - **Immutable Transactions**: Makes previously recorded transactions difficult to modify without detection, supporting reliable recordkeeping and accountability.
  - **Transaction Validation**: Transactions are validated according to the rules of the smart contract and the consensus mechanism of the underlying blockchain network.
  - **Cryptographic Verification:** Uses cryptographic hashing to help verify data integrity unauthorized changes where integrity checks are implemented.

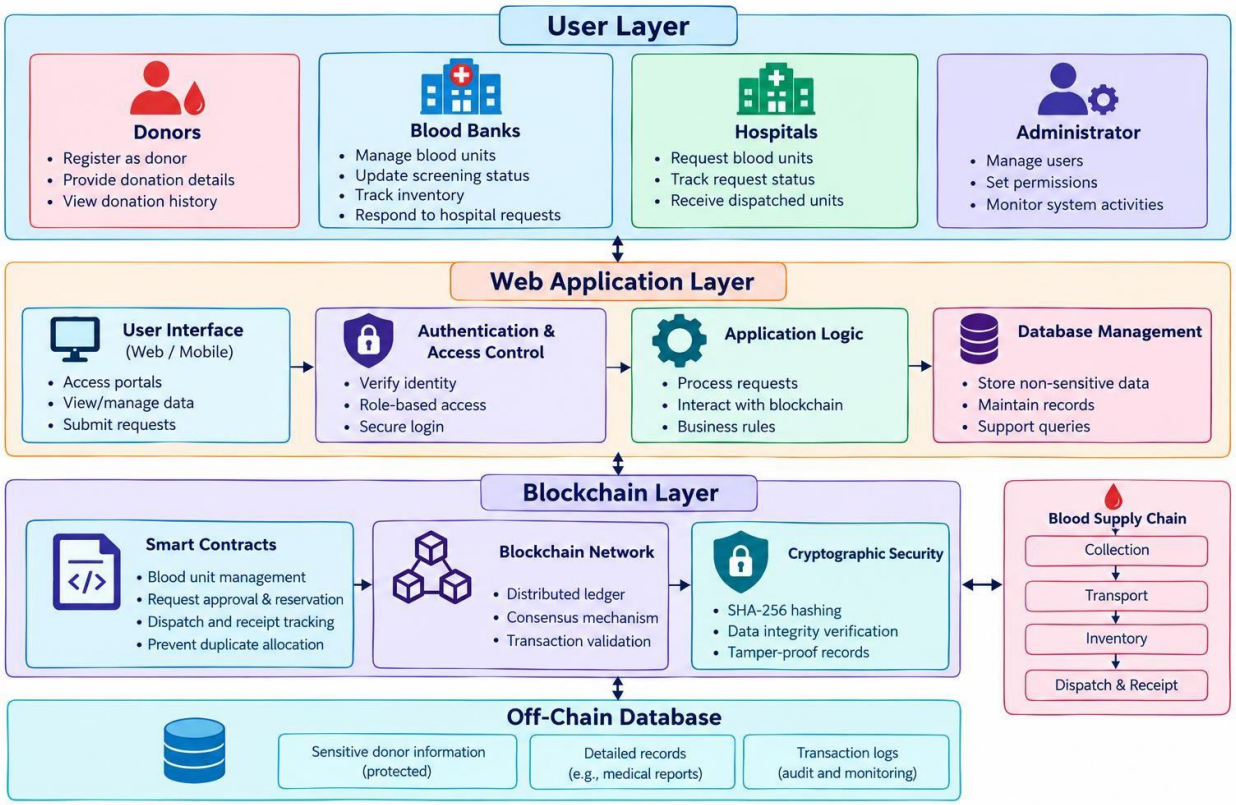


Figure 4.1: Architecture of Blockchain based Blood Bank Management System

5.IMPLEMENTATION

5.1. Setting Up the Blockchain Environment

To implement the Blockchain-Based Blood Bank Management System, the development environment is configured using Solidity and the Ethereum-compatible blockchain environment. The smart contract is developed using Solidity version 0.8.20, with Hardhat serving as the development and testing framework. Ganache provides a local blockchain for deploying and testing smart contracts, while Ethers.js enables communication between the application and the blockchain.

The system uses React.js for the frontend, Node.js and Express.js for backend services, and MongoDB for storing application data. The blockchain maintains important blood supply chain records, while sensitive donor information is kept off-chain to support privacy. After configuring the environment, the smart contract is deployed and integrated with the backend and frontend. [1]

5.2. Defining Smart Contracts

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.20;
contract BloodSupplyChain {
    address public admin;

    enum UnitStatus {
        Collected, PendingScreening, Approved, Available,
        Reserved, Dispatched, Received, Transfused,
        Quarantined, Expired, Discarded
    }

    enum ScreeningStatus {
        Pending, Passed, Failed
    }
}
```

These enumerations represent the different stages of a blood unit and its laboratory screening status.

5.2.1. Registering Blood Banks and Hospitals

The smart contract allows the administrator to register blood banks and hospitals by recording their names, license numbers, authorization status, and registration status. Only authorized organizations can perform the operations permitted by their roles. This helps prevent unauthorized participation and improves accountability throughout the blood supply chain.

```
struct BloodBank {
        string name;
        string licenseNumber;
        bool isAuthorized;
        bool exists;
    }
    struct Hospital {
        string name;
        string licenseNumber;
        bool isAuthorized;
        bool exists;
    }
```

5.2.2 Registering Blood Units

Each blood unit is registered with a unique unit ID, blood group, component type, collection timestamp, expiry timestamp, screening status, and data hash. The system checks that the unit ID is valid and has not already been registered. It also ensures that the expiry timestamp is later than the collection timestamp. After registration, the unit's details and registration event are recorded, providing a traceable starting point for its journey through the supply chain.

```
function registerBloodUnit() external {
    string calldata _unitId,
    string calldata _bloodGroup,
    string calldata _componentType,
    uint256 _collectionTimestamp,
    uint256 _expiryTimestamp,
    string calldata _dataHash
}
```

5.2.3 Updating Blood Screening Status

The system records the screening result of each blood unit. If screening is passed, the unit becomes available for reservation. If screening fails, the unit is marked as quarantined. If screening is still pending, the unit remains under screening and is not available for allocation. These checks help prevent unsuitable blood units from progressing through the normal supply chain. Actual laboratory testing must still be performed by qualified medical personnel.

```
if (_screeningStatus == ScreeningStatus.Passed) {
    unit.status = UnitStatus.Available;
    _recordEvent(_unitId, "ScreeningPassed",
       UnitStatus.Available, msg.sender, _dataHash);
    emit ScreeningStatusUpdated(_unitId, _screeningStatus,
       UnitStatus.Available, msg.sender, _dataHash);
 } else if (_screeningStatus == ScreeningStatus.Failed) {
    unit.status = UnitStatus.Quarantined;
    _recordEvent(_unitId, "ScreeningFailed",
       UnitStatus.Quarantined, msg.sender, _dataHash);
    emit ScreeningStatusUpdated(_unitId, _screeningStatus,
       UnitStatus.Quarantined, msg.sender, _dataHash);
 } else {
    unit.status = UnitStatus.PendingScreening;
    _recordEvent(_unitId, "ScreeningPending",
       UnitStatus.PendingScreening, msg.sender, _dataHash);
    emit ScreeningStatusUpdated(_unitId, _screeningStatus,
       UnitStatus.PendingScreening, msg.sender, _dataHash);
 }
```

5.2.4 Approving Hospital Blood Requests

When a hospital requests blood, the system checks the request ID, hospital authorization, blood group, component type, screening result, and availability of the selected unit. If all conditions are satisfied, the blood unit is reserved for the designated hospital. The system records the request details and links the unit to the request. Reserving the unit immediately helps prevent duplicate allocation to multiple hospitals.

```
require(bytes(_requestId).length > 0, "Request ID empty");
require(!bloodRequests[_requestId].exists, "Request already exists");
require(hospitals[_hospital].isAuthorized, "Hospital is not authorized");
BloodUnit storage unit = bloodUnits[_unitId];
require(unit.status == UnitStatus.Available, "Unit is not available");
require(unit.screeningStatus == ScreeningStatus.Passed,"Unit has not passed screening");
require(keccak256(bytes(unit.bloodGroup)) == keccak256(bytes(_bloodGroup)),"Blood
group mismatch");
require(keccak256(bytes(unit.componentType)) == keccak256(bytes(_componentType)),
  "Component type mismatch");
require(unit.bloodBank == msg.sender || msg.sender == admin,"Unit belongs to another
blood bank");
require(unit.currentCustodian == unit.bloodBank,"Unit is not held by its blood bank");
unit.status = UnitStatus.Reserved; unit.hospital = _hospital; unit.requestId = _requestId;
```

5.2.5 Dispatching Blood Units

After a request is approved, the blood bank dispatches the reserved blood unit to the designated hospital. Before recording dispatch, the system checks the participant's authorization, the unit's current custodian, its reservation status, and the destination hospital. The dispatch record includes shipment information and a data hash. This provides a traceable record of the transfer from the blood bank to the hospital.

```
function dispatchBloodUnit(
    string calldata _unitId,
    address _destination,
    string calldata _shipmentRef,
    string calldata _dataHash
) external onlyAuthorizedParticipant unitExists(_unitId) notExpired(_unitId) {
    BloodUnit storage unit = bloodUnits[_unitId];
    require(unit.currentCustodian == msg.sender || msg.sender == admin,
        "Not custodian");
    require(unit.status == UnitStatus.Reserved,
        "Unit must be reserved before dispatch");
    require(_destination == unit.hospital,
        "Destination does not match assigned hospital");
    require(bytes(unit.requestId).length > 0,
        "Unit is not linked to a request");
    unit.status = UnitStatus.Dispatched;
    unit.dataHash = _dataHash;
    _recordEvent(_unitId, "Dispatched",
        UnitStatus.Dispatched, msg.sender, _dataHash);
    emit BloodUnitDispatched(_unitId, msg.sender,
        _destination, _shipmentRef, _dataHash);
}
```

5.2.6 Receiving Blood Units

When the blood unit reaches the hospital, its receipt is recorded in the smart contract. The system verifies that the unit has been dispatched and that the receiving organization is the designated hospital or an authorized administrator. The unit's status and custody information are then updated, and a receipt event is recorded. This creates a verifiable record of the handover between the blood bank and the hospital.

```
function receiveBloodUnit(
    string calldata _unitId,
    string calldata _dataHash
) external onlyAuthorizedHospital unitExists(_unitId) {
    BloodUnit storage unit = bloodUnits[_unitId];

    require(unit.status == UnitStatus.Dispatched, "Not dispatched");
    require(unit.hospital == msg.sender || msg.sender == admin,
        "Caller is not designated hospital");

    unit.status = UnitStatus.Received;
    unit.currentCustodian =
        (msg.sender == admin && unit.hospital != address(0))
        ? unit.hospital : msg.sender;
    unit.dataHash = _dataHash;

    _recordEvent(_unitId, "Received", UnitStatus.Received,
        unit.currentCustodian, _dataHash);
    emit BloodUnitReceived(_unitId, unit.currentCustodian,
        block.timestamp, _dataHash);
}
```

5.2.7 Recording Emergency Blood Transfers

The system supports recording emergency blood transfers between hospitals. It stores information such as the transfer ID, blood unit ID, requesting hospital, supplying hospital, urgency, and transfer status. Before creating a transfer record, the system checks that the transfer ID has not already been used, that the unit exists, that it has been received, and that it has not expired. This functionality helps hospitals coordinate urgent blood requirements while maintaining a record of the transfer.

```
function createEmergencyTransfer(
    string calldata _transferId,
    string calldata _unitId,
    address _supplyingHospital,
    string calldata _urgency
) external onlyAuthorizedHospital {
    require(emergencyTransfers[_transferId].requestTimestamp == 0,
        "Transfer ID exists");
    BloodUnit storage unit = bloodUnits[_unitId];
    require(unit.exists, "Unit does not exist");
    require(unit.status == UnitStatus.Received, "Unit not available");
    require(block.timestamp < unit.expiryTimestamp, "Unit expired");
    emergencyTransfers[_transferId] = EmergencyTransfer({
        transferId: _transferId,
        unitId: _unitId,
        requestingHospital: msg.sender,
        supplyingHospital: _supplyingHospital,
        requestTimestamp: block.timestamp,
        completionTimestamp: 0,
        isCompleted: false,
        urgency: _urgency});
    emit EmergencyRequestCreated(_transferId, _unitId, msg.sender,
        _supplyingHospital, _urgency);}
```

5.2.8 Verifying Data Integrity

The smart contract includes a data integrity verification mechanism that compares a stored data hash with an expected hash. If the values match, the function returns a successful result; otherwise, the values do not match. This helps identify differences between recorded hash values and the values supplied for verification. However, this mechanism does not independently verify laboratory results or guarantee the physical safety and quality of the blood.

```
function verifyDataIntegrity(
    string calldata _unitId,
    string calldata _expectedHash
) external view returns (bool) {
    if (!bloodUnits[_unitId].exists) return false;

    return keccak256(bytes(bloodUnits[_unitId].dataHash))
        == keccak256(bytes(_expectedHash));
}
```

5.2.9 Maintaining Blood Supply Chain Records

The smart contract records events for important operations, including blood unit registration, screening updates, reservation, dispatch, receipt, transfusion, expiry, and disposal. These records allow authorized participants to review the history of a blood unit and trace its movement through the supply chain. Maintaining a consistent event history improves transparency, supports auditing, and helps identify the stage at which a problem occurred.

5.3 Deploying and Integrating with a Frontend

After the smart contract is developed, it is compiled and tested using Hardhat. The contract is then deployed to the local Ganache blockchain for development and testing. The deployment process makes the contract address and ABI available to the backend so that it can interact with the deployed contract.

The frontend is developed using React.js, while the backend uses Node.js and Express.js. Ethers.js connects the backend to the blockchain, and MongoDB stores application data that should not be placed directly on-chain, including sensitive donor information.

The application provides role-based interfaces for administrators, blood bank staff, hospital staff, and donors. Through these interfaces, users can perform the operations permitted by their roles, including blood unit registration, inventory tracking, hospital requests, dispatch, receipt, and emergency transfers.

The application can be tested locally before any deployment to a public blockchain network. A public deployment should be performed only after appropriate testing, access-control review, and security assessment.

5.4 Workflow:

The workflow begins with the registration and authorization of blood banks and hospitals. Authorized staff register blood units with details such as blood group, component type, collection time, and expiry time.

Each unit undergoes laboratory screening. Approved units become available for reservation, while failed units are quarantined and excluded from allocation.

When an authorized hospital requests blood, the blood bank selects a suitable unit. The smart contract approves the request and reserves the unit. The blood bank dispatches it, and the hospital confirms receipt.

The system records the unit's final status, such as transfused, expired, or discarded. Blockchain events and data hashes support traceability and integrity checks. Emergency transfers between authorized hospitals can also be recorded.

This workflow improves inventory tracking, reduces duplicate allocation, and provides transparent records of blood movement. However, laboratory testing, proper storage, and safe transportation remain essential for blood safety.

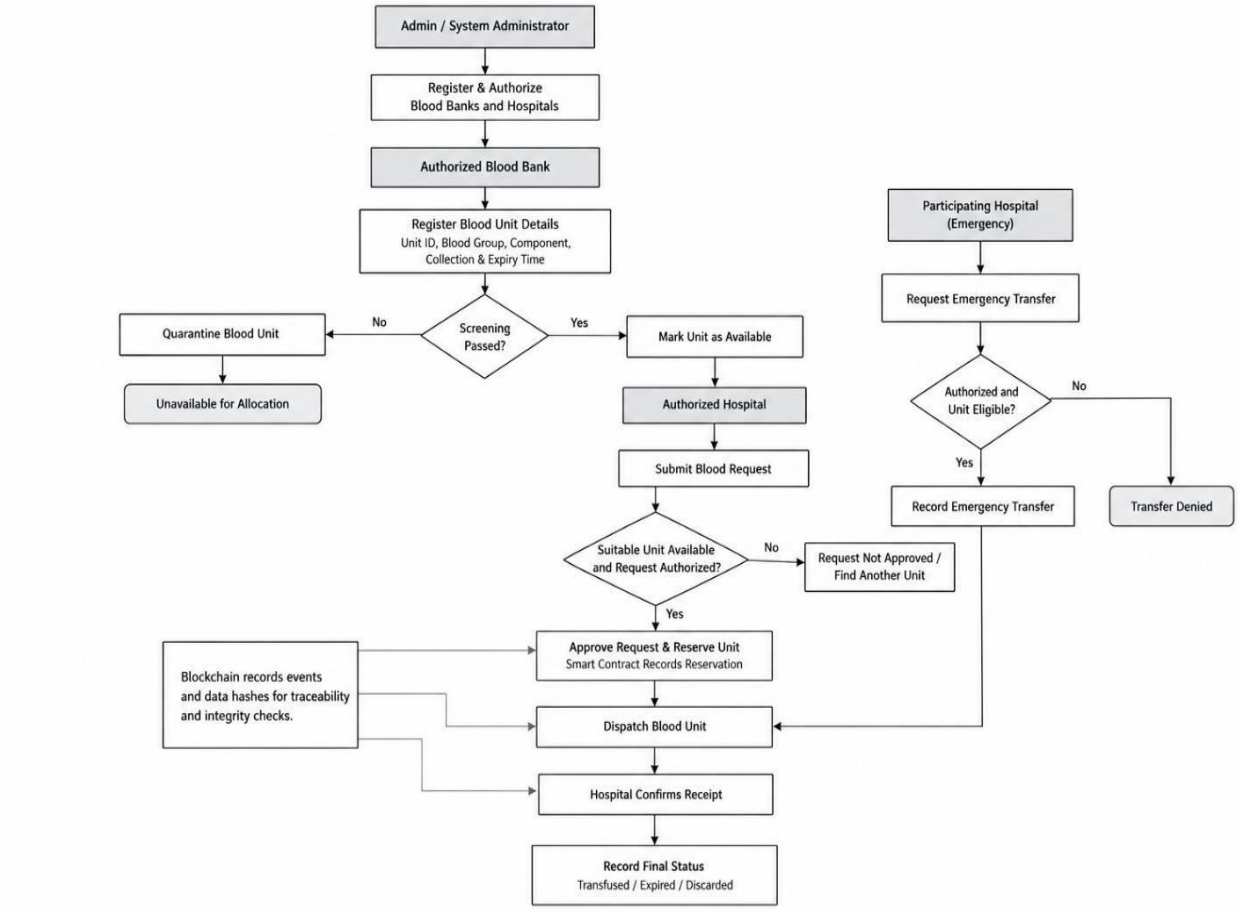


Figure 5.1: Workflow of Blockchain based Blood Bank Management System

6. BENEFITS

Blockchain technology enhances the Blood Bank Management System by improving data security, transparency, traceability, and coordination between blood banks and hospitals. Traditional blood bank systems often depend on centralized databases and manual records, which may lead to data loss, unauthorized modifications, duplicate allocation, and delays in blood distribution. By using blockchain and smart contracts, the proposed system maintains traceable transaction records and supports secure blood supply chain management. The following are the key benefits:

6.1. Enhanced Security and Immutability

Blockchain maintains records of blood unit registration, screening status, reservation, dispatch, and receipt. Once transactions are recorded on the blockchain, modifying previous records becomes difficult without detection. This helps protect blood supply chain information against unauthorized changes and improves the reliability of records.

6.2. Improved Blood Unit Traceability

The system helps track blood units throughout the supply chain, from registration and screening to reservation, dispatch, and hospital receipt. Each unit has a unique identification number, making it easier for authorized participants to identify its status and review its transaction history.

6.3. Transparency and Auditability

Blockchain records important transactions and events, allowing authorized participants to verify the history of blood units. Blood banks and hospitals can review relevant records to understand the movement and status of units. This improves accountability and supports auditing of blood distribution activities.

6.4. Prevention of Duplicate Blood Unit Allocation

The smart contract checks the availability and current status of a blood unit before reservation. Once a unit is reserved, it is no longer treated as generally available for another allocation. This helps reduce duplicate allocation and improves the accuracy of blood inventory management.

6.5. Improved Inventory Management

The system records blood unit details, screening results, and status changes. This helps blood banks identify available, reserved, dispatched, received, and quarantined units. Better inventory visibility can support timely decisions, reduce unnecessary delays, and help minimize wastage due to expiry.

6.6. Automated Processing Through Smart Contracts

Smart contracts enforce predefined conditions for operations such as blood unit registration, screening status updates, hospital request approval, reservation, dispatch, and receipt confirmation. This reduces dependence on manual record updates and helps apply the same rules consistently.

6.7. Faster Hospital Blood Request Processing

Hospitals can submit blood requests, while the smart contract checks relevant information such as blood group, component type, screening status, and availability. Automating these checks can reduce administrative delays and help blood banks respond more efficiently to hospital requirements.

6.8. Improved Data Integrity

It includes data-hash verification to compare a stored hash value with an expected value. This can help detect discrepancies in recorded data when the application correctly generates and verifies the hashes. It supports integrity checking but does not independently guarantee the accuracy of laboratory results or blood quality.

6.9. Better Coordination During Emergencies

The system includes emergency transfer functionality to record transfers between hospitals. Such records can help participating organizations coordinate blood availability and review emergency transfer details. This can support better resource utilization when blood is urgently required.

6.10. Increased Accountability and Trust

Authorized participants can review recorded transactions and identify the stages through which a blood unit has passed. This improves accountability among blood banks and hospitals and helps reduce disputes regarding registration, reservation, dispatch, and receipt.

6.11. Reduced Paperwork and Administrative Effort

Maintaining digital transaction records can reduce reliance on paper-based documentation and repetitive manual recordkeeping. Smart contract automation and traceable records can improve operational efficiency, although staff are still required for laboratory testing, physical handling, transportation, and verification.

In conclusion, blockchain improves blood bank security, traceability, and inventory management, while medical testing and safe blood handling remain essential.

7. CHALLENGES

Although blockchain-based blood bank management offers several benefits, it also has challenges and limitations that must be considered during implementation. The following are the key challenges associated with the proposed system.

7.1. Scalability Issues

As the number of blood units, hospital requests, and transactions increases, the blockchain network may experience delays and higher transaction costs. This can affect the speed of processing blood requests, especially during emergencies.

7.2. Data Privacy and Security

Blood bank systems handle sensitive donor and medical information. Storing such information directly on a blockchain may create privacy concerns because recorded data can be difficult to remove. Therefore, sensitive information must be protected through appropriate access controls and secure off-chain storage.

7.3. Dependence on Accurate Data Entry

Blockchain protects recorded transactions from unauthorized modification, but it cannot guarantee that the original information is correct. Incorrect blood group details, screening results, or expiry dates entered by authorized users may affect the reliability of the system.

7.4. Internet Connectivity and Availability

The system depends on network connectivity to access blockchain services and submit transactions. Poor internet connectivity or blockchain network disruptions may delay blood unit updates and hospital request processing.

7.5. Integration and User Adoption

Integrating blockchain with existing blood bank and hospital systems may require additional development, training, and maintenance. Users must also understand how to operate the system correctly to ensure effective adoption.

Despite these challenges, the Blockchain-Based Blood Bank Management System can improve traceability, transparency, and coordination. Proper security measures, accurate data entry, user training, and a suitable blockchain network are essential for effective implementation.

8. CONCLUSION

The Blockchain-Based Blood Bank Management System addresses challenges in traditional blood bank management, including inaccurate records, limited traceability, duplicate allocation, and delays in blood distribution. By using blockchain technology and smart contracts, the proposed system improves data security, transparency, and coordination between blood banks and hospitals. It supports blood unit registration, screening status tracking, hospital requests, reservation, dispatch, and receipt confirmation.

The system provides a structured approach to maintaining reliable transaction records and improving blood supply chain management. However, laboratory testing, blood compatibility verification, proper storage, and safe transportation remain essential for ensuring blood safety.

8.1. Future Outlook for Enhancements

In the future, the system can be enhanced by integrating Internet of Things (IoT) sensors to monitor blood storage temperature and transportation conditions. Artificial intelligence (AI) can help predict blood demand, identify shortages, and reduce wastage due to expiry.

Additionally, mobile applications and real-time notifications can improve communication between blood banks and hospitals. Stronger identity verification, improved privacy protection, and integration with existing hospital management systems can further enhance security and usability. Scalability improvements can also support a larger number of users and transactions.

The system can also be enhanced with GPS-based tracking to monitor blood transportation and provide updates on delivery status. Automated alerts can notify authorized personnel about expiring blood units, low inventory levels, and delays during transportation. These features can help blood banks take timely action and improve the availability of blood during emergencies.

Furthermore, advanced data analytics can be used to study blood usage patterns and improve inventory planning. Integration with regional blood bank networks could help hospitals locate available blood units across different locations. Regular security audits and smart contract testing can strengthen the reliability of the system and reduce potential vulnerabilities.

By incorporating these enhancements, the Blockchain-Based Blood Bank Management System can become more efficient, secure, reliable, and suitable for wider adoption in healthcare organizations.

9. SDG's ADDRESSED

The Blockchain-Based Blood Bank Management System supports the United Nations Sustainable Development Goals (SDGs) by improving healthcare coordination, promoting technological innovation, and reducing the wastage of valuable medical resources. The following three SDGs are addressed by the proposed system:

SDG 3: Good Health and Well-Being

The Blockchain-Based Blood Bank Management System primarily supports **SDG 3: Good Health and Well-Being**, which aims to ensure healthy lives and promote well-being for people of all ages. Blood is essential for treating patients during surgeries, accidents, childbirth complications, and other medical emergencies. Therefore, maintaining an efficient and reliable blood supply is an important part of healthcare services.

The proposed system improves coordination between blood banks and hospitals by maintaining records of blood units, blood groups, component types, screening results, and expiry dates. It helps authorized hospital staff identify suitable available units and process blood requests more efficiently. During emergencies, the system can support faster communication and coordination between participating hospitals and blood banks, helping reduce delays in blood distribution.

Blockchain technology provides traceable records of blood unit registration, reservation, dispatch, and receipt. Smart contracts enforce predefined rules and help prevent duplicate allocation of blood units. These features improve inventory visibility, accountability, and transparency throughout the blood supply chain. The system can also help blood banks monitor inventory and identify units approaching their expiry dates, reducing avoidable wastage.

By improving blood inventory management and coordination, the system can support timely access to blood and contribute to better healthcare services. However, blockchain alone cannot guarantee blood safety. Proper laboratory screening, blood compatibility testing, appropriate storage conditions, and safe transportation must always be maintained to protect patients.

By integrating blockchain technology, the Blockchain-Based Blood Bank Management System supports SDG, promoting secure blood tracking, improved healthcare coordination, and responsible blood inventory management.

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11. APPENDIX A

https://drive.google.com/drive/folders/1bbol0R2a0misWfRKNDNwaj910wxISHVG?usp=sharing

https://github.com/BorraHemaPriya18/blockchain-blood-bank